Supporting Information for *Evaluating multi-season occupancy models with autocorrelation fitted to heterogeneous datasets*

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G Results of multi-season site occupancy model for the remaining four butterfly species

This file shows the maps and tables (coefficients, Rhat and Prior-Posterior Overlap) from the occupancy analyses for four butterfly species whose results were not shown in the main text: the false ringlet *Coenonympha oedippus* (Fabricius, 1787), the marsh fritillary *Euphydryas aurinia* (Rottemburg, 1775), the small copper *Lycaena phlaeas* (Linnaeus, 1761), and the meadow brown *Maniola jurtina* (Linnaeus, 1758). The analyses were performed at the scale of the greater Bordeaux area, and used weakly informative priors for the spatial decay parameter ϕ in the occupancy models (as described in the main text).

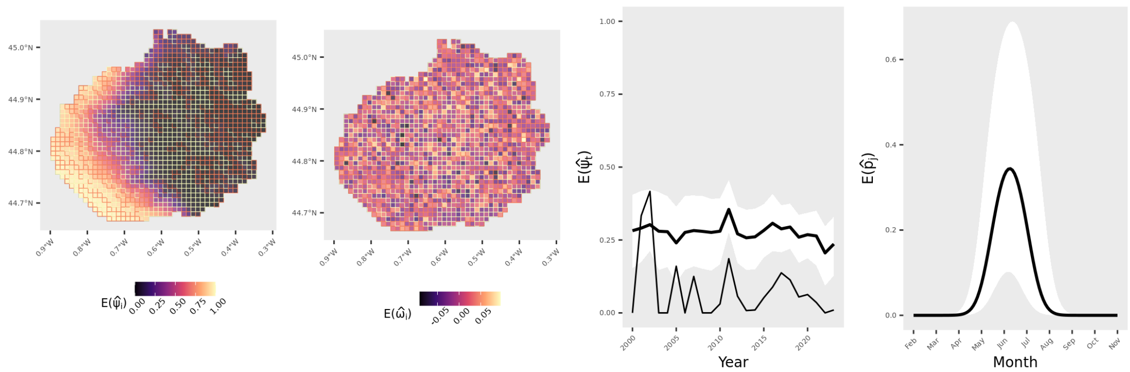

Fig. G.1: Distribution of the false ringlet *Coenonympha oedippus* in the greater Bordeaux area, Southwest France. Results for the model with NNGP=15 and weakly informative prior for ϕ. Values plotted in the occupancy map (left) were calculated as $\mathbb{E}(\hat{\psi}_i) = \frac{1}{T}\sum_{t=1}^{T} \hat{\psi}_{it}$, and refer to in-sample estimates for sites with data (cells with yellow borders) and out-of-sample predictions for missing cells (those with red borders). The spatial random effect $\mathbb{E}(\hat{\omega}_i)$ was also derived from in-sample estimates and out-of-sample predictions from the model. Trend plots show the proportion of occupied sites over the 24 years of data, calculated as $\mathbb{E}(\hat{\psi}_t) = \frac{1}{I}\sum_{i=1}^{I} \hat{\psi}_{it}$. The naive occupancy trend (number of sites with detection divided by the total number of sampled sites per year) is depicted by the thinner line. Detection probability over months (secondary periods) was also derived from in-sample predictions. The black line depicts the average detection trend, and the white bands depict the upper and lower bounds of the 95% Credible Interval. No 1×1 km cell was 100% covered by water (the maximum was 93%). Parameter values derived from 1,200 posterior distribution draws.

Table G.1: Estimated parameters of the multi-season site-occupancy model with spatial and temporal random effects fitted to data of the false ringlet *Coenonympha oedippus* (Fabricius, 1787). Results for models with NNGP=15 spatial neighbors, and weak or informative ϕ priors. PPO: Prior-Posterior Overlap. PPO: Prior-Posterior Overlap. Abbreviation of covariates = LAT: Latitude, LON: Longitude, ELEV= Elevation, LAND= non-water land cover, GRASS: Cover of natural grasslands, OBS: Number of observers, MON: survey month. The mean, 95% Credible Intervals, and Rhat were calculated using the 1,200 posterior distribution draws of each parameter.

Parameter	Mean	LCI	UCI	Rhat	Overlap
β_0	-4.623	-6.220	-3.069	1.058	6.7
β_{LAT}	0.227	-0.692	1.124	1.086	45.2
β_{LAT^2}	1.571	0.877	2.357	1.189	27.3
β_{LON}	-3.781	-4.933	-2.716	1.088	8.7
β_{ELEV}	2.108	1.043	3.278	1.016	29.1
$\beta_{NON-WATER}$	-0.941	-3.757	1.861	1.146	75.0
$\beta_{NON-WATER^2}$	-1.555	-4.551	0.756	2.621	60.8
β_{URBAN}	-0.693	-2.792	0.552	1.173	62.7
σ^2	1.154	0.354	2.767	1.570	83.2
ϕ	32.184	4.555	59.016	1.003	8.0
σ_T^2	0.986	0.233	3.108	1.013	86.6
ρ	0.072	-0.311	0.420	1.022	43.7
α_0	-0.721	-2.324	0.780	1.161	61.0
$\alpha_{NON-WATER}$	0.403	-2.798	3.241	1.399	87.6
α_{LAT}	0.382	0.039	0.707	1.032	19.1
α_{OBS}	0.064	-0.048	0.189	1.007	8.5
α_{MON}	-1.733	-2.732	-0.743	1.005	33.2
α_{MON^2}	-9.937	-12.261	-7.929	1.019	0.0

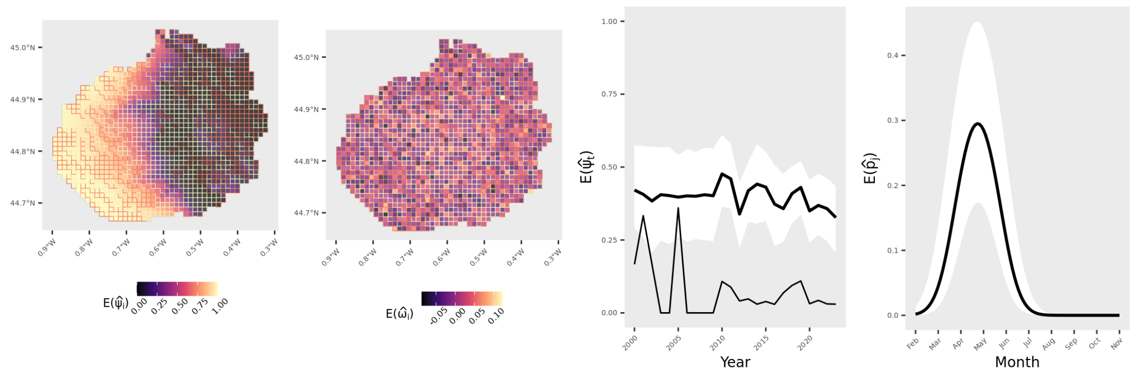

Fig. G.2: Distribution of the marsh fritillary *Euphydryas aurinia* in the greater Bordeaux area, Southwest France. Results for the model with NNGP=15 and weakly informative prior for ϕ. Values plotted in the occupancy map (left) were calculated as $\mathbb{E}(\hat{\psi}_i) = \frac{1}{T}\sum_{t=1}^{T} \hat{\psi}_{it}$, and refer to in-sample estimates for sites with data (cells with yellow borders) and out-of-sample predictions for missing cells (those with red borders). The spatial random effect $\mathbb{E}(\hat{\omega}_i)$ was also derived from in-sample estimates and out-of-sample predictions from the model. Trend plots show the proportion of occupied sites over the 24 years of data, calculated as $\mathbb{E}(\hat{\psi}_t) = \frac{1}{I}\sum_{i=1}^{I} \hat{\psi}_{it}$. The naive occupancy trend (number of sites with detection divided by the total number of sampled sites per year) is depicted by the thinner line. Detection probability over months (secondary periods) was also derived from in-sample predictions. The black line depicts the average detection trend, and the white bands depict the upper and lower bounds of the 95% Credible Interval. No 1×1 km cell was 100% covered by water (the maximum was 93%). Parameter values derived from 1,200 posterior distribution draws.

Table G.2: Estimated parameters of the multi-season site-occupancy model with spatial and temporal random effects fitted to data of the marsh fritillary *Euphydryas aurinia* (Rottemburg, 1775). Results for models with 15 spatial neighbors, and weak or informative ϕ priors. PPO: Prior-Posterior Overlap. PPO: Prior-Posterior Overlap. Abbreviation of covariates = LAT: Latitude, LON: Longitude, ELEV= Elevation, LAND= non-water land cover, GRASS: Cover of natural grasslands, OBS: Number of observers, MON: survey month. The mean, 95% Credible Intervals, and Rhat were calculated using the 1,200 posterior distribution draws of each parameter.

Parameter	Mean	LCI	UCI	Rhat	Overlap
β_0	-2.122	-3.512	-0.821	1.011	30.8
β_{LAT}	0.337	-0.494	1.227	1.019	44.8
β_{LAT^2}	1.194	0.585	1.938	1.217	28.5
β_{LON}	-3.637	-4.952	-2.417	1.257	9.3
β_{ELEV}	1.767	0.770	2.916	1.339	31.5
$\beta_{NON-WATER}$	-0.675	-2.706	1.285	1.008	72.9
$\beta_{NON-WATER^2}$	-0.677	-1.681	0.203	1.022	44.8
β_{URBAN}	-1.135	-4.232	0.227	2.635	60.2
σ^2	1.297	0.278	3.760	2.671	90.7
ϕ	31.397	4.437	58.805	1.015	8.5
σ_T^2	1.360	0.262	4.194	1.037	69.3
ρ	0.061	-0.317	0.422	1.042	44.8
α_0	-3.091	-4.271	-1.964	1.002	15.3
$\alpha_{NON-WATER}$	-1.100	-2.665	0.174	1.011	52.8
α_{LAT}	0.014	-0.270	0.299	1.014	17.6
α_{OBS}	0.057	-0.035	0.155	1.020	6.8
α_{MON}	-7.264	-8.947	-5.699	1.060	0.4
α_{MON^2}	-5.943	-7.383	-4.581	1.036	1.3

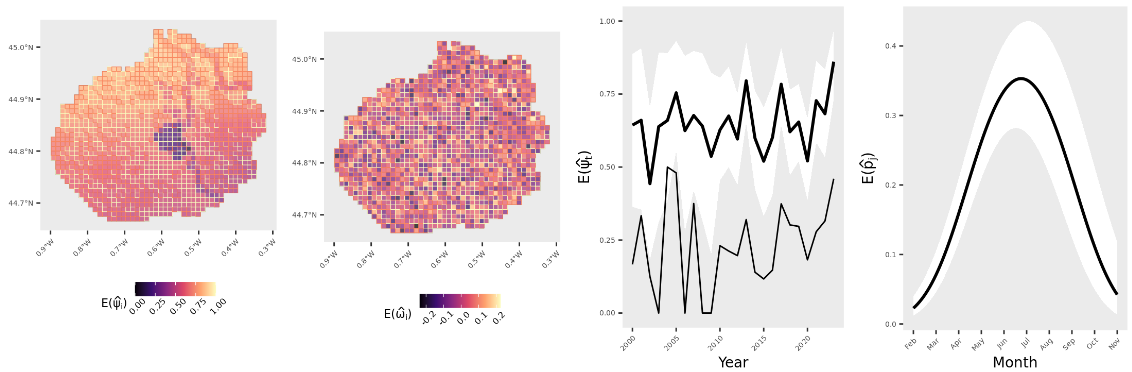

Fig. G.3: Distribution of the small copper *Lycaena phlaeas* in the greater Bordeaux area, Southwest France. Results for the model with NNGP=15 and weakly informative prior for ϕ. Values plotted in the occupancy map (left) were calculated as $\mathbb{E}(\hat{\psi}_i) = \frac{1}{T}\sum_{t=1}^{T} \hat{\psi}_{it}$, and refer to in-sample estimates for sites with data (cells with yellow borders) and out-of-sample predictions for missing cells (those with red borders). The spatial random effect $\mathbb{E}(\hat{\omega}_i)$ was also derived from in-sample estimates and out-of-sample predictions from the model. Trend plots show the proportion of occupied sites over the 24 years of data, calculated as $\mathbb{E}(\hat{\psi}_t) = \frac{1}{I}\sum_{i=1}^{I} \hat{\psi}_{it}$. The naive occupancy trend (number of sites with detection divided by the total number of sampled sites per year) is depicted by the thinner line. Detection probability over months (secondary periods) was also derived from in-sample predictions. The black line depicts the average detection trend, and the white bands depict the upper and lower bounds of the 95% Credible Interval. No 1×1 km cell was 100% covered by water (the maximum was 93%). Parameter values derived from 1,200 posterior distribution draws.

Table G.3: Estimated parameters of the multi-season site-occupancy model with spatial and temporal random effects fitted to data of the small copper *Lycaena phlaeas* (Linnaeus, 1761). Results for models with 15 spatial neighbors, and weak or informative ϕ priors. PPO: Prior-Posterior Overlap. PPO: Prior-Posterior Overlap. Abbreviation of covariates = LAT: Latitude, LON: Longitude, ELEV= Elevation, LAND= non-water land cover, GRASS: Cover of natural grasslands, OBS: Number of observers, MON: survey month. The mean, 95% Credible Intervals, and Rhat were calculated using the 1,200 posterior distribution draws of each parameter.

Parameter	Mean	LCI	UCI	Rhat	Overlap
β_0	1.142	-0.045	2.338	1.068	45.9
β_{LAT}	1.052	0.103	1.971	1.190	40.3
β_{LAT^2}	0.258	-0.475	0.970	1.106	36.6
β_{LON}	-0.356	-1.160	0.260	1.038	36.9
β_{ELEV}	-0.190	-0.807	0.476	1.015	33.8
$\beta_{NON-WATER}$	0.354	-1.526	2.136	1.014	69.8
$\beta_{NON-WATER^2}$	-0.051	-0.745	0.616	1.033	35.7
β_{URBAN}	-0.613	-1.025	-0.256	1.019	22.9
σ^2	5.961	1.809	12.356	1.026	18.1
ϕ	31.658	4.257	59.072	1.000	8.7
σ_T^2	1.878	0.505	5.083	1.057	49.4
ρ	0.006	-0.387	0.376	1.001	45.0
α_0	-0.614	-0.935	-0.296	1.042	18.0
$\alpha_{NON-WATER}$	0.008	-0.139	0.161	1.002	10.0
α_{LAT}	-0.201	-0.369	-0.028	1.036	10.8
α_{OBS}	0.026	0.000	0.052	1.030	2.3
α_{MON}	0.204	0.036	0.373	1.009	12.0
α_{MON^2}	-1.152	-1.386	-0.935	1.011	11.5

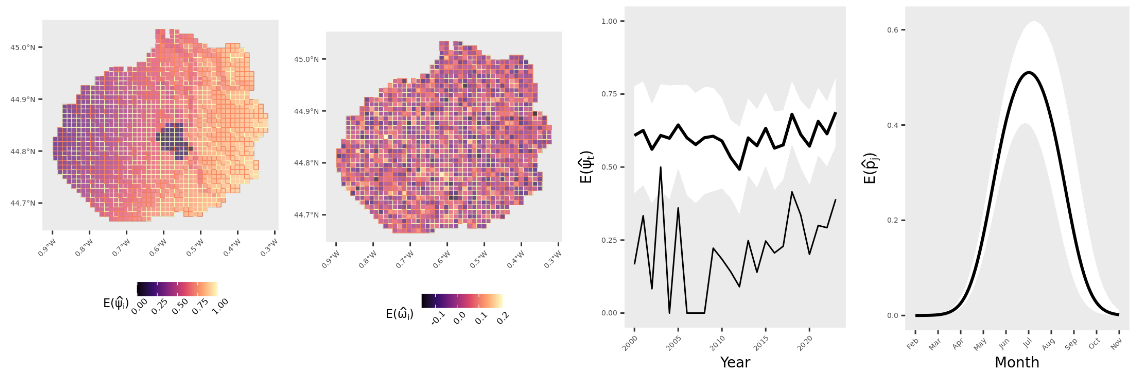

Fig. G.4: Distribution of the meadow brown *Maniola jurtina* in the greater Bordeaux area, Southwest France. Results for the model with NNGP=15 and weakly informative prior for ϕ. Values plotted in the occupancy map (left) were calculated as $\mathbb{E}(\hat{\psi}_i) = \frac{1}{T}\sum_{t=1}^{T} \ \hat{\psi}_{it}$, and refer to in-sample estimates for sites with data (cells with yellow borders) and out-of-sample predictions for missing cells (those with red borders). The spatial random effect $\mathbb{E}(\hat{\omega}_i)$ was also derived from in-sample estimates and out-of-sample predictions from the model. Trend plots show the proportion of occupied sites over the 24 years of data, calculated as $\mathbb{E}(\hat{\psi}_t) = \frac{1}{I}\sum_{i=1}^{I} \ \hat{\psi}_{it}$. The naive occupancy trend (number of sites with detection divided by the total number of sampled sites per year) is depicted by the thinner line. Detection probability over months (secondary periods) was also derived from in-sample predictions. The black line depicts the average detection trend, and the white bands depict the upper and lower bounds of the 95% Credible Interval. No 1×1 km cell was 100% covered by water (the maximum was 93%). Parameter values derived from 1,200 posterior distribution draws.

Table G.4: Estimated parameters of the multi-season site-occupancy model with spatial and temporal random effects fitted to data of the meadow brown *Maniola jurtina* (Linnaeus, 1758). Results for models with 15 spatial neighbors, and weak or informative ϕ priors. PPO: Prior-Posterior Overlap. PPO: Prior-Posterior Overlap. Abbreviation of covariates = LAT: Latitude, LON: Longitude, ELEV= Elevation, LAND= non-water land cover, GRASS: Cover of natural grasslands, OBS: Number of observers, MON: survey month. The mean, 95% Credible Intervals, and Rhat were calculated using the 1,200 posterior distribution draws of each parameter.

Parameter	Mean	LCI	UCI	Rhat	Overlap
β_0	0.632	-0.192	1.538	1.058	44.7
β_{LAT}	-0.178	-0.905	0.475	1.072	36.8
β_{LAT^2}	0.188	-0.237	0.671	1.032	28.1
β_{LON}	1.305	0.744	2.041	1.025	30.2
β_{ELEV}	0.097	-0.511	0.705	1.037	34.1
$\beta_{NON-WATER}$	0.057	-1.467	1.454	1.042	65.0
$\beta_{NON-WATER^2}$	-0.061	-0.629	0.461	1.048	31.3
β_{URBAN}	-0.649	-1.030	-0.354	1.056	18.5
σ^2	3.338	1.186	7.623	1.074	32.6
ϕ	31.594	4.697	58.407	1.004	8.5
σ_T^2	0.468	0.158	1.261	1.001	70.0
ρ	0.031	-0.363	0.412	1.019	45.1
α_0	-0.065	-0.444	0.288	1.012	22.0
$\alpha_{NON-WATER}$	-0.094	-0.235	0.042	1.020	10.0
α_{LAT}	0.100	-0.081	0.288	1.015	12.1
α_{OBS}	0.045	0.017	0.077	1.001	2.5
α_{MON}	1.188	0.919	1.488	1.002	14.6
α_{MON^2}	-3.344	-3.822	-2.863	1.008	5.8